Platelets

Dr. Jehad Alhmoud

Platelets (Thrombocytes)

- **Platelets**, also called **thrombocytes**, are a component of blood whose function (along with the coagulation factors) is in hemostasis (prevent bleeding and help stop it when occurs).
- The 2^{nd} most numerous blood cells.
- They are cellular fragments that are derived from the megakaryocytes.
- They are cells consisting of granular cytoplasm with a membrane but no nuclear material.
- Have a diameter of 2.5 μm with a mean platelet volume (MPV) of 8-10 fL.
- Circulating, resting platelets are biconvex.

- The normal peripheral blood platelet count is 150 to 400 X 10^9/L.

- This count represents only two-thirds (⅔) of available platelets because the spleen sequesters an additional one-third (⅓).

- Sequestered platelets are immediately available in times of demand—for example, in acute inflammation, after an injury, or after major surgery.

- In hypersplenism or splenomegaly, increased sequestration may cause a relative thrombocytopenia.

Platelets Ultrastructure

- Platelets, although anucleate, are complex and metabolically active.

- The plasma membrane of platelets resembles any biological membrane a bilayer composed of proteins and lipids.

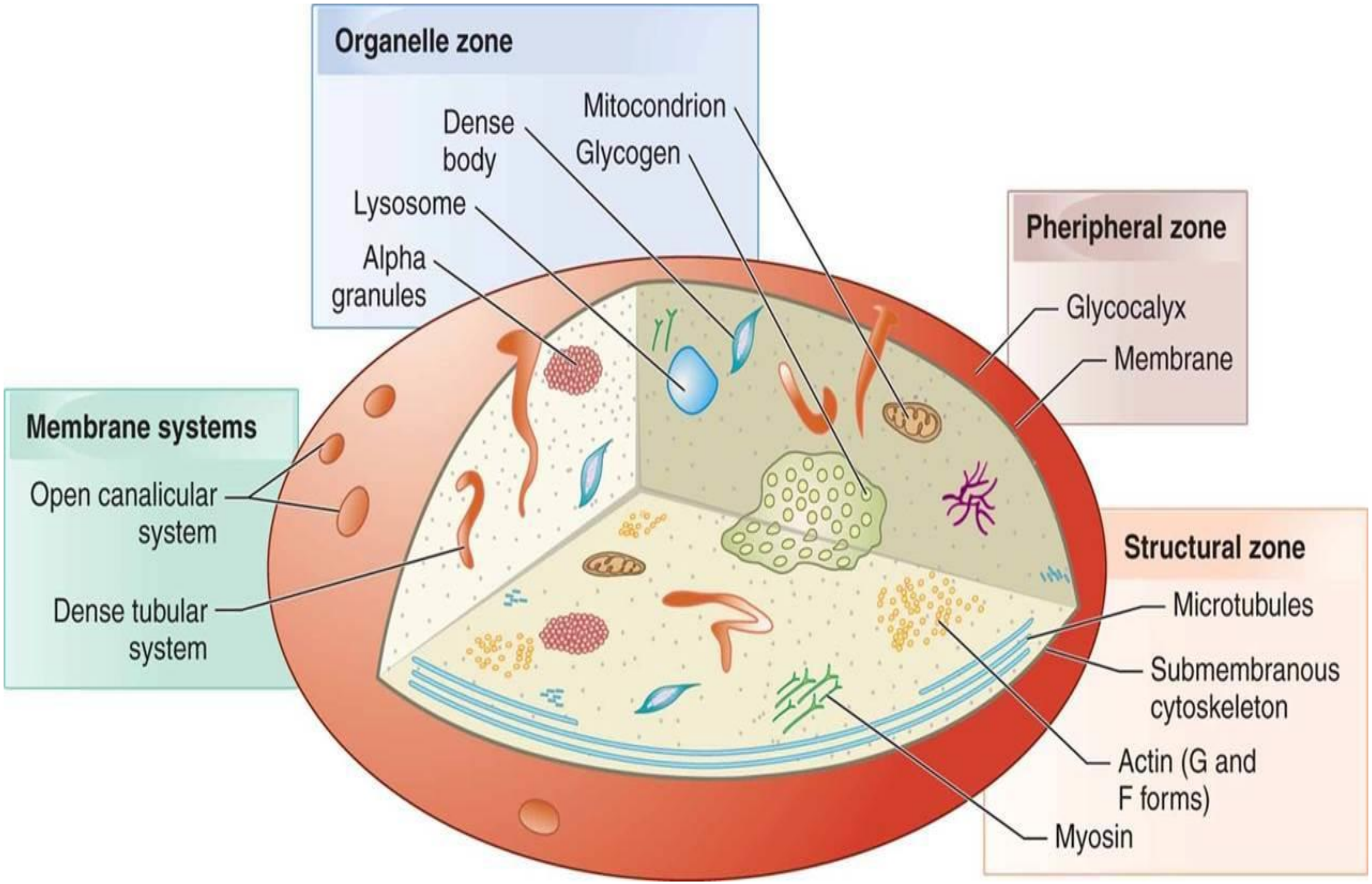
- Structurally the platelet can be divided into four zones, from peripheral to innermost:

1. Peripheral zone - is rich in glycoproteins required for platelet adhesion, activation, and aggregation. For example, GPIb/IX/X; GPVI; GPIIb/IIIa.

2. Sol-gel zone - is rich in microtubules and microfilaments, allowing the platelets to maintain their discoid shape. Platelets contain actin filaments, myosin, and other contractile proteins, which help them retain their shape, squeeze out the contents of the cytoplasmic granules, and allow platelet plugs to contract.

3. Organelle zone - is rich in platelet granules.

4. Membranous zone - contains membranes derived from megakaryocytic endoplasmic reticulum.


Organelle zone
Mitocondrion
Dense body
Glycogen
Lysosome
Alpha granules
Pheripheral zone
Glycocalyx
Membrane
Membrane systems
Open canalicular system
Dense tubular system
Structural zone
Microtubules
Submembranous cytoskeleton
Actin (G and F forms)
Myosin

Platelet Receptors

- Platelets have a variety of surface glycoproteins, some of which act as receptors for von Willebrand factor (vWF), fibrinogen, or other adhesive proteins.

- Many platelet receptors consist of complexes of two or more glycoproteins.

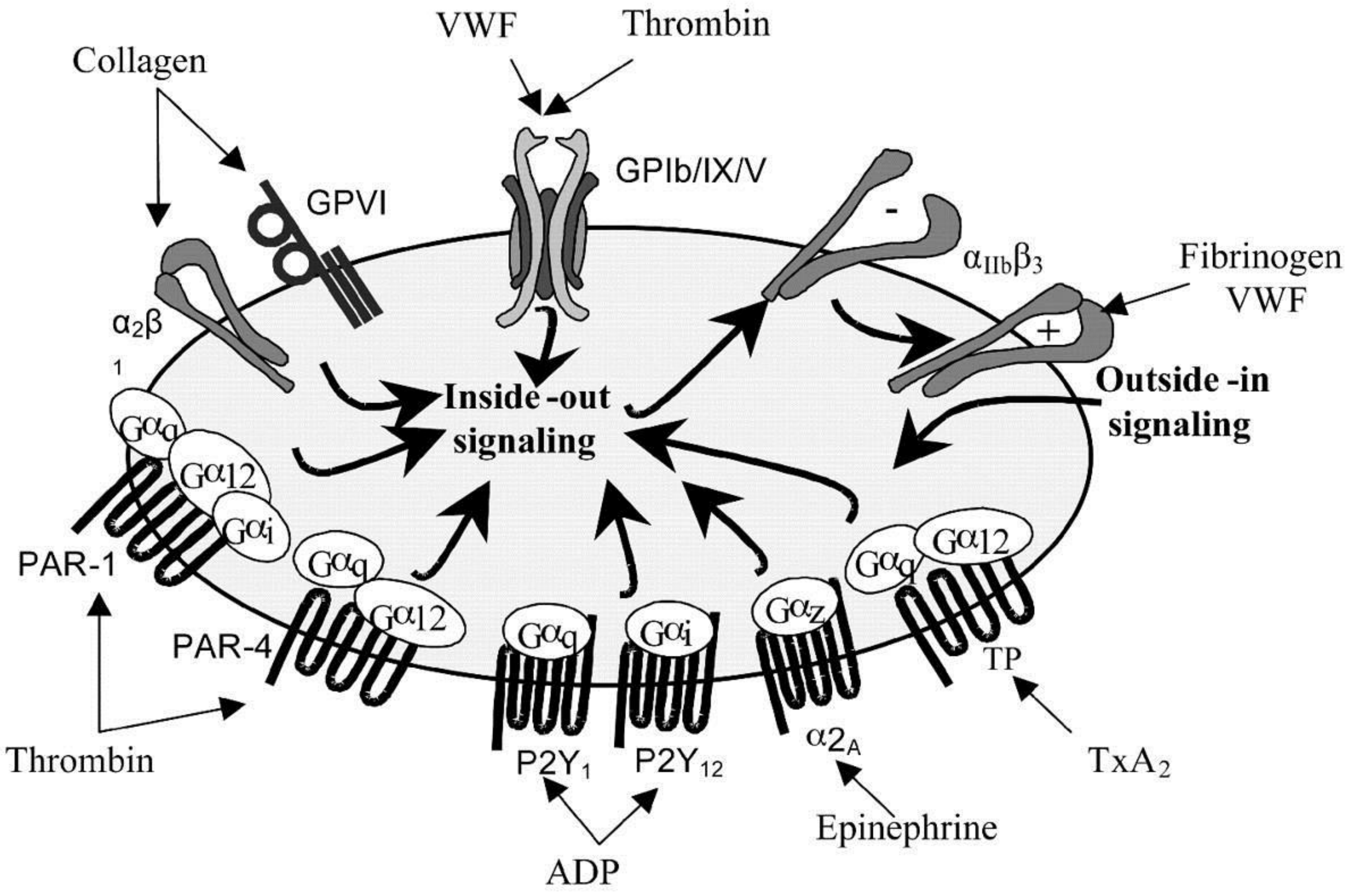
- The most important platelet receptors are the following:

- **GP Ib/IX/V**: The platelet receptor for **vWF** which mediates the capture of platelets to the injured vascular wall and **thrombin**.

- **GP IIb/IIIa**: The platelet receptor for **fibrinogen and vWF**. GP IIb-IIIa exists on the resting platelet in a low-affinity or inactive form. After the platelet is activated by initial adhesion, the GP IIb-IIIa undergoes a conformational change to a high-affinity form, and additional IIb-IIIa is transferred from the interior to the exterior of the platelet.

- **Glycoprotein Ia/IIa, VI:** is the receptor for **collagen.**

- There are also receptors on platelets for ADP, epinephrine and thromboxane A**2** that activate platelets and for prostacyclin which inhibits platelets aggregation.



Platelet Receptors

Role of Platelets in Hemostasis

It involves three processes:

1. **Adhesion:** the adherence and attachment of platelets to the injured endothelium.

2. **Activation and Secretion:** once the platelet is activated, it secretes granular contents of alpha granules and dense bodies.

3. **Aggregation:** is the accumulation of many platelets at the site of injury.

Platelet adhesion

- Is the property by which platelets bind non-platelet surfaces such as subendothelial collagen.

- When the endothelial layer of a blood vessel is disrupted, subendothelial collagen is exposed to the circulation.

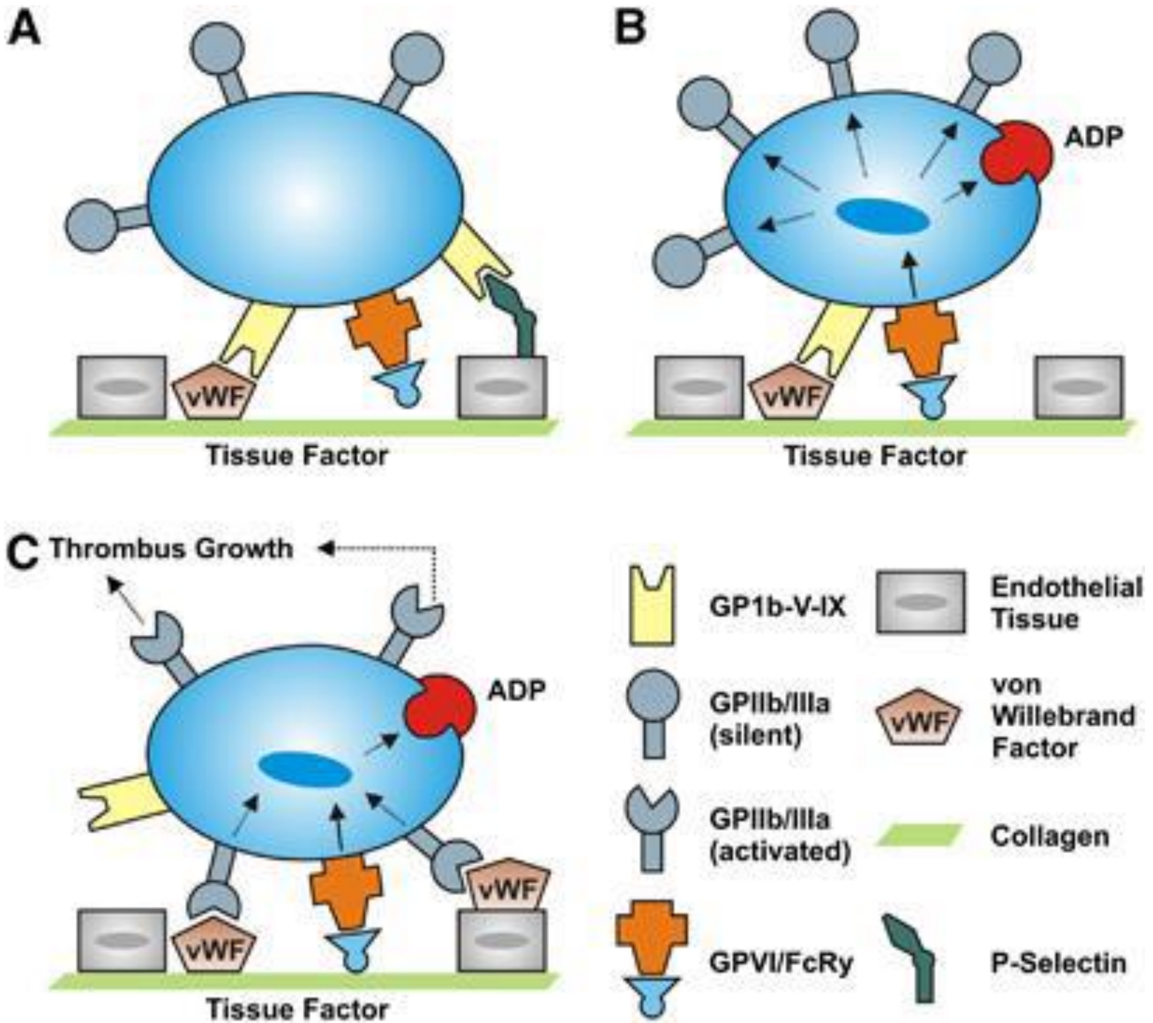
- Platelets bind to the exposed collagen through the GP Ia/IIa and GP VI.

- Damaged endothelial cells release VWF from cytoplasmic storage organelles.

- VWF binds to subendothelial collagen and GP Ib/IX/V (the vWF receptor) on platelet surfaces, resulting in platelet adhesion and activation.

- GP IIb/IIIa (the fibrinogen receptor) on the platelet surface is converted from a low-affinity to a high-affinity form, and additional GP IIb/IIIa is brought to the platelet surface.

- Platelet adhesion is a reversible process.



Distinct steps of platelet adhesion, activation, and aggregation at the activated endothelium.

A. The initial adhesion of platelets (tethering) is mediated by the binding of the glycoprotein (GP)Ib-V–IX receptor complex to the A1 domain of the von Willebrand factor (VWF) on endothelial cells. Additionally, binding to P-Selectin can enhance platelet recruitment to the intact vessel wall.

B. In the second step, interactions between GPVI and collagen stabilize the thrombus. Moreover, it comes to cellular activation with the secretion of platelet agonists (e.g., adenosine diphosphate, ADP) and transformation of the GPIIb/IIIa receptors to a state with high affinity.

C. The common final pathway of the platelet activation via the GPIIb/GPIIIa (integrin–fibrinogen) pathway culminates in an irreversible platelet aggregation and subsequent thrombus growth.

Platelet Aggregation

- Aggregation of platelets is a process in which more platelets accumulate at the site of injury. (is the property by which platelets bind to one another).

- Platelet aggregation occurs with the aid of certain substances such as ADP and thromboxane A_2**.**

- ADP is a substance that attracts more platelets and thromboxane A_2 promotes vasoconstrictor and platelet aggregation and degranulation.

- Aggregation is a part of primary hemostasis, and its irreversible endpoint is the "white clot" or platelet plug, a clot composed primarily of platelets and VWF.

- Aggregation occurs as a result of conformational change of GPIIb/IIIa receptor, which allows these receptors to bind with vWF or fibrinogen.

- Fibrinogen acts as a bridge that binds between platelets to form platelet aggregation.

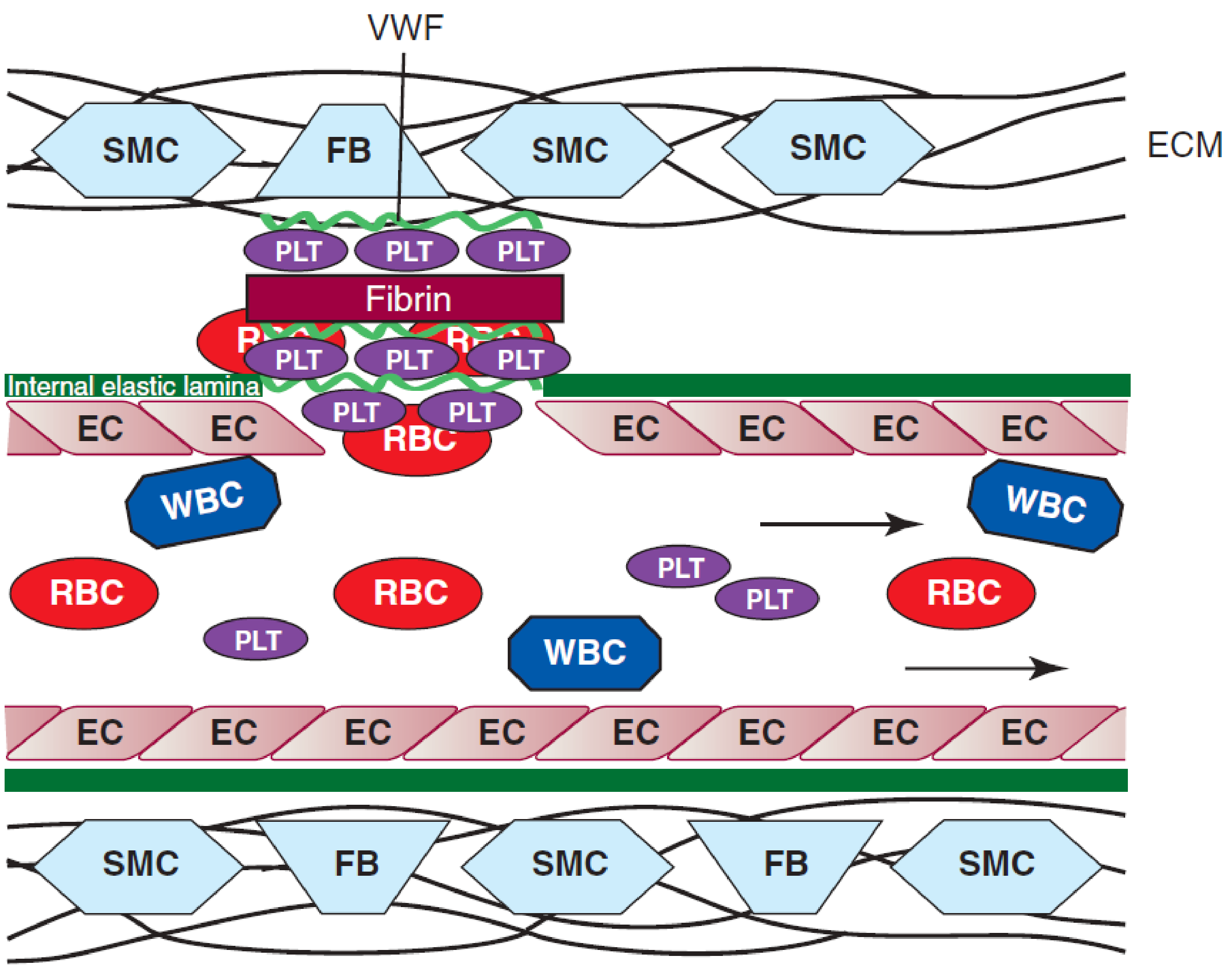
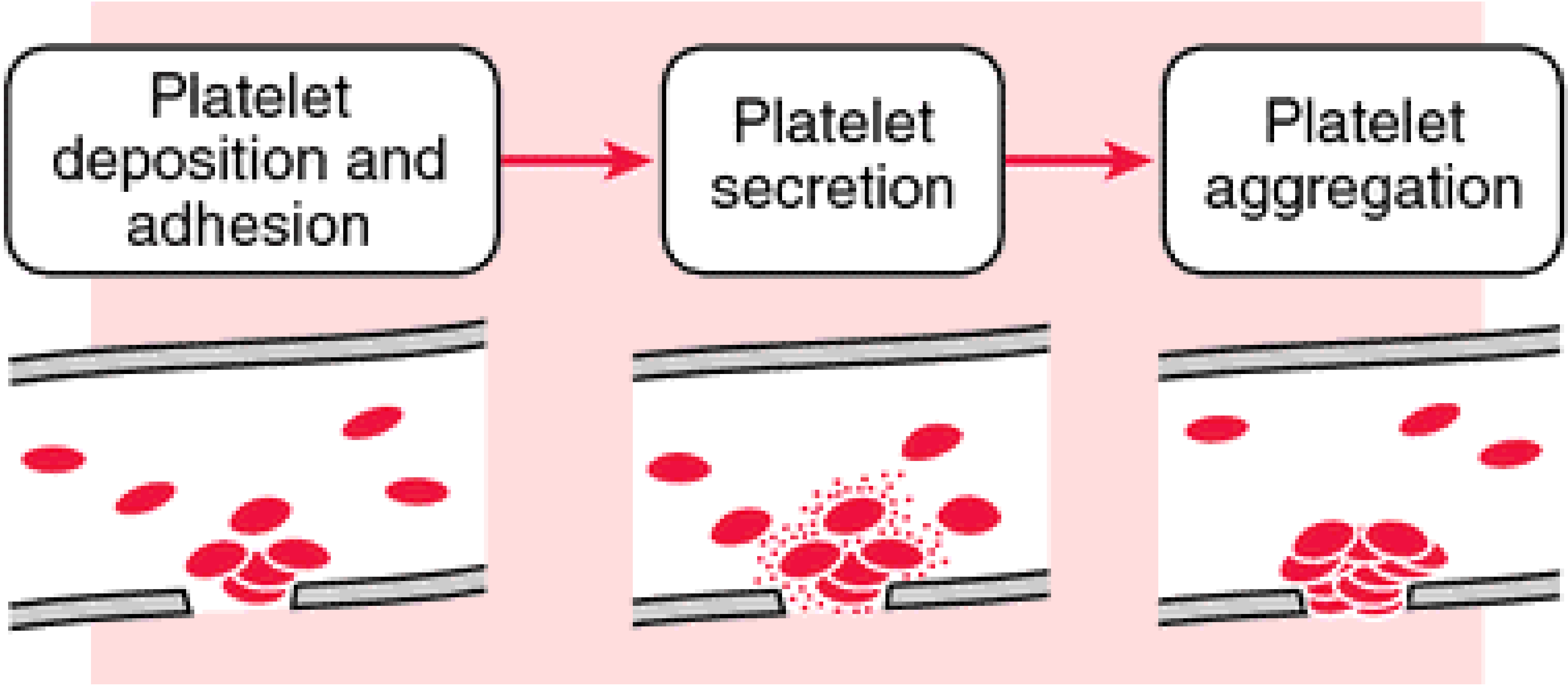
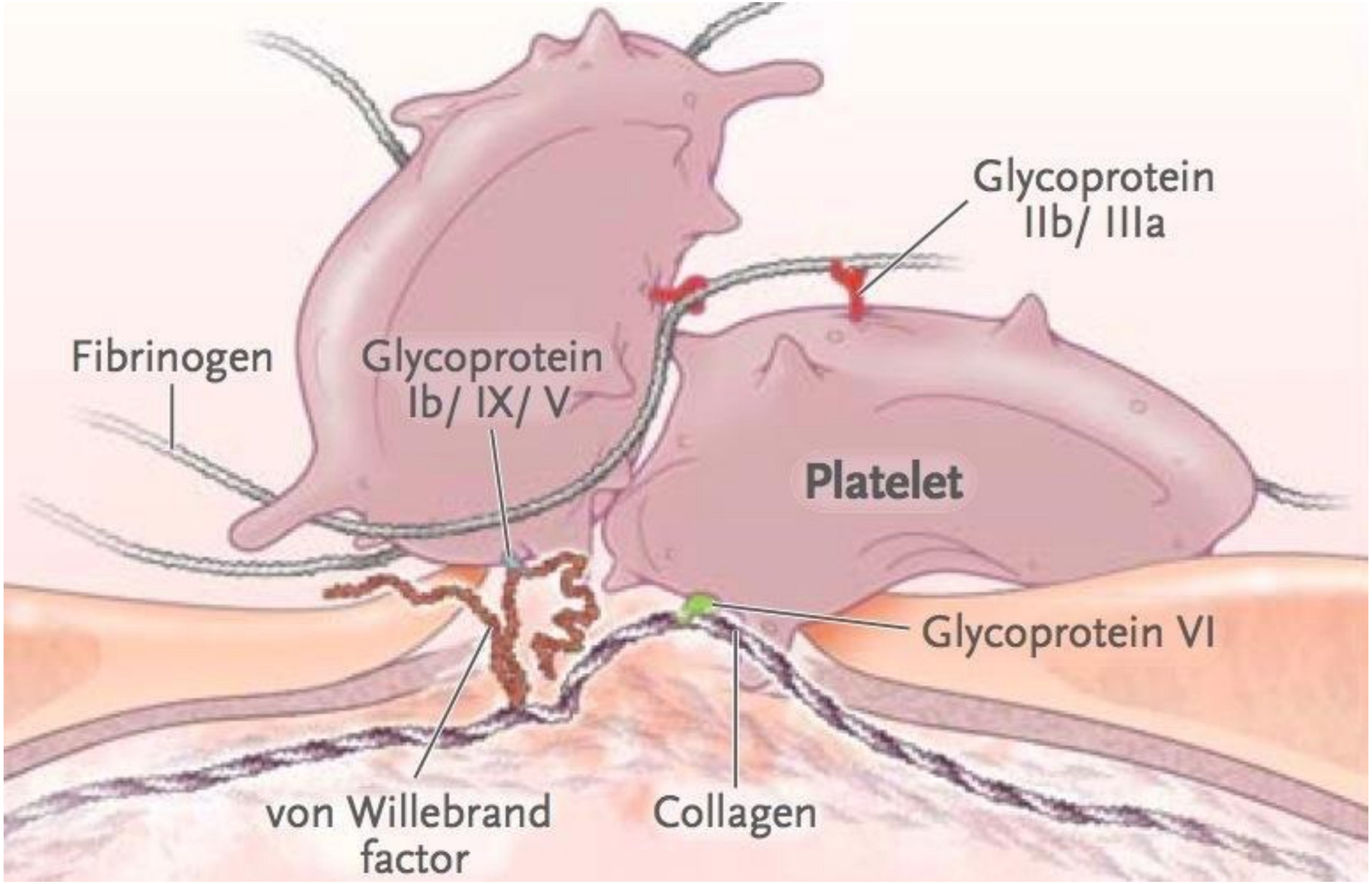


Figure 37-5 Platelet aggregation. In veins and venules, the bulky "red clot" consists of platelets *(PLT)*, von Willebrand factor *(VWF)*, fibrin, and red blood cells *(RBC)*. Though a protective mechanism, the red clot may occlude the vessel, causing venous thromboembolic disease. *EC,* Endothelial cell; *ECM,* extracellular matrix; *FB,* fibroblast; *WBC,* white blood cell; *SMC,* smooth muscle cell; *lines* indicate collagen.


Platelet deposition and adhesion
Platelet secretion
Platelet aggregation



Glycoprotein
IIb/ IIIa
Fibrinogen
Glycoprotein
Ib/ IX/ V
Platelet
Glycoprotein VI
von Willebrand
factor
Collagen